25 June 1973
SUPERSEDING
MIL-B-43308A
24 March 1969

MILITARY SPECIFICATION

BINDING MACHINE, PLASTIC BINDING ELEMENT, AND PUNCHING MACHINE, PAPER

1. SCOPE

1.1 Scope. This specification covers bench mounted punching and binding machines used for plastic strip binding of pamphlets, booklets, catalogs, and similar publications (see 6.1).

1.2 Classification. The machines shall be of the following types, as specified (see 6.2).

Type I - Punching Machine, 16-Inch Capacity, Electric Motor Driven.
Type II - Binding Machine, 16-Inch Capacity, Hand Lever Operated.
Type III- Punching and Binding Machine, 12-Inch Capacity, Hand Lever Operated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

FSC 3610

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
* MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
* MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL)

Standard No. 114 - Office Appliances and Business Equipment.

Electrical Appliance and Utilization Equipment List.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 333 Pfingsten Road, Northbrook, Illinois 60062; 1285 Walt Whitman Road, Melville, New York 11746; or 1655 Scott Boulevard, Santa Clara, California 95090.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.4).

* 3.2 Standard product. The machines delivered under this specification, shall be the manufacturers commercial product, except for any changes necessary to comply with specification requirements. All like items furnished under any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Standards compliance.

3.3.1 Safety standard. Type I machines shall conform to the requirements of UL Standard No. 114, for cord connected appliances.

* 3.3.2 Certification. Prior to approval of the first article, if one is submitted or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the type I machines he proposes to furnish under this specification meet the applicable requirements of UL Standard 114. Acceptable evidence of meeting the requirements of UL shall be the UL Label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the type I machines have been tested and conform to UL Standard No. 114.

3.4 Materials. Except as otherwise specified herein, materials and components shall be as used in the manufacturer's standard commercial punching and binding machines.

3.5 Design and construction. The machines shall be designed for bench mounting and for operation from a sitting or standing position.

3.5.1 Punching machines. The punching machines shall be designed to punch a single row of 1/8 by 5/16 inch (± 0.005 inch) holes on 9/16 inch (± 1/64 inch non-cumulative) centers. Side and back guides shall be incorporated with manual adjustment, to insure the required margin adjacent to the punched holes for various diameter binding elements from 3/16 or 7/32 inch to 1-1/8 inch diameter, and equal margins at each end of any size sheet within the limits of the machine. Manually operated punch releases shall also be incorporated to deactivate and reactivate any number of hole punches as may be desired, within the limits of the machine, without the need of tools or removal of any components or parts of the machine. The operating mechanism shall be designed so that all of the desired holes are punched in one cycle of the operating mechanism. A removable chip receptacle shall be provided to catch punched-out waste.

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3.5.2 Binding machines. The binding machines shall be hand lever operated and designed for installing plastic strip binding elements in collated material having punched holes conforming to the hole size, hole spacing, and location of holes specified in 3.5.1. The machines shall be capable of installing binding elements of various sizes from 3/16 or 7/32 inch inside diameter thru 1-1/8 inch inside diameter, in collated material of pile heights commensurate with the binding element used. An adjustable stop shall be provided to control the required spread of the binding element for installation in the collated material.

3.5.3 Type I, punching machine. Type I punching machine shall conform to 3.5.1, shall be electric motor driven, and shall be capable of punching 28 holes within 16 inches, for each cycle of the punching mechanism (see 4.6.1). In addition, the lift capacity shall be not less than 22 sheets of paper having an average thickness of 0.003 inch per sheet. The punching action shall be induced in response to paper pile pressure applied to a switch incorporated in the side guide or, at the supplier's option, to pressure applied to a foot operated switch. The punching cycle shall be nonrepeating, and shall require reapplication of pressure to the switch for each succeeding cycle of the punching mechanism.

3.5.3.1 Electrical characteristics. The type I machines shall operate on a 60-Hertz, 120 volt power supply.

3.5.3.2 Electromagnetic compatibility. When specified (see 6.2), the type I machines shall be designed and equipped for electromagnetic compatibility in accordance with IIB of MIL-STD-461 (see 4.6.3).

3.5.4 Type II, binding machine. The type II binding machine shall conform to 3.5.2 and shall have a capacity to install a binding element up to 16 inches long in collated material for each cycle of the operating element.

3.5.5 Type III, punching and binding machine. The type III machine shall be a hand lever operated, single unit, combination machine, consisting of a punching section and a binding section. The punching section shall conform to 3.5.1 and shall be capable of punching 21 holes within 12 inches for each cycle of the operating mechanism (see 4.6.1). In addition, the punching section lift capacity shall be not less than 15 sheets of paper having an average thickness of 0.003 inch per sheet. The binding section shall conform to 3.5.2 and shall have a capacity to install binding elements up to 12 inches long in material previously punched in the punching section.

3.6 Performance characteristics. The punching machines shall punch clean, sharp, distortion free, equally spaced holes, with no localized marginal or size variation between holes, when tested in accordance with 4.6.1. When the binding machines are tested in accordance with 4.6.2, the installed binding element shall be completely closed, with a binding element ring through each hole of collated material, and free of visible

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evidence of distortion, tears, or cutting of the binding element or collated material. Individual pages of the bound material shall have complete freedom of movement when turned over, page after page, and without tearing or pulling away from the binding.

3.7 Finish. The applied finish shall be in accordance with the manufacturer's commercial practice.

3.8 Marking for identification. Marking for identification shall conform to the unit marking requirements of MIL-STD-130.

* 3.9 Workmanship. All components and assemblies of the machines shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and corners, except when corners are required.

3.9.1 Metal fabrication. Metal used in the fabrication of machines shall provide for original quality surface finish and shall be free from kinks and unspecified binds. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size and shape. Painted and plated surfaces shall be uniformly coated and free from runs, teardrops, sags, grit, foreign inclusions, blisters, peeling or other defects indicative of faulty surface preparation or application of finishes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article. When a first article is required, the machine shall be tested as specified in 4.6.1 through 4.6.3, as applicable. Failure of any test shall be cause for rejection of the first article.

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* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in

4.4.1 End item inspection. The inspection lot shall be all the machines of one type offered for inspection at one time.

4.4.1.1 Visual examination. The machines shall be examined for the defects in table I. The inspection level shall be II with an AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examination	Defect	Classification Major	Classification Minor
Finish	Run, teardrop, sag, grit, or foreign inclusion		X
	Blister, peel, or flake		X
Construction and workmanship	Burr, snag, or hazardous projection	X	
	Fastening device loose		X
	Improper assembly	X	
	Welds or soldered connections faulty, when applicable	X	
	Any component or assembly missing	X	
	Not designed for bench mounting	X	
	Not type machine required	X	
Identification marking	Missing, illegible, incorrect, or improperly applied		X

* 4.4.1.2 Dimensional examination. The machines shall be examined for compliance with dimensions specified. Any dimension not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.4.1.3 End item testing. When a first article is not required, the first produced machine shall be tested as specified in 4.6.1 through 4.6.3, as applicable, and failure of any test shall be cause for rejection of the first produced machine. Each sample machine shall be tested as specified in 4.6.1 through 4.6.3, as applicable, and failure to pass a test shall be recorded as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects expressed as defects per hundred units.

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4.4.2 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2. The AQL shall be 4.0, expressed in terms of defects per hundred units.

* 4.5 Certificate examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

TABLE II. Examination of preparation for delivery

Examination	Defect
Markings (exterior and interior)	Missing; illegible; incorrect; improper size, location, sequence, or method of application.
Material	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components, such as cushioning, blocking, and bracing inadequate or improperly fixed.
Contents	Number per container is more or less than specified.

4.6 Test methods. The machines shall be set up for operation in accordance with the manufacturer's operating instruction.

4.6.1 Punching machines. The punching machines shall be tested using paper having an average thickness of 0.003 inch per sheet. Sheet size for testing type I machines shall be 11 by 17 inches, and the lift shall be 22 sheets. Sheet size for testing the punching section of type III machines shall be 8-1/2 by 13 inches, and the lift shall be 15 sheets. With all of the hole punches activated, the appropriate size and lift of paper shall be placed under the punches and against the back and side guides. The machine shall then be cycled through one complete punching stroke. The punched pile shall then be removed and examined as follows:

(a) The size of the punched holes, and the center distance between the end holes, in a randomly selected sheet, shall be measured to determine compliance with dimensional requirements in 3.5.1.

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(b) The number of holes punched in each sheet shall be counted to determine compliance with 3.5.3 or 3.5.5, as applicable.

(c) The punched holes shall be examined to determine compliance with the quality requirements of 3.6.

(d) A randomly selected sheet shall be folded, end to end, with the edge adjacent to the punched holes precisely aligned, and the end holes overlayed. The overlayed holes shall then be examined for uniformity of margin, hole size and hole spacing as specified in 3.6.

The action of the punch release mechanism shall be tested by deactivating all hole punches, placing an appropriate size sheet of paper in punching position, and cycling the machine. Each hole punch shall then be activated in turn, and a separate sheet of paper punched for each hole punch activated. Examination of the sheets shall be made after each cycle of the machine to determine compliance with the applicable requirement in 3.5.1.

4.6.2 Binding machines. The binding machines shall be tested using offset book paper having an average thickness of 0.003 inch per sheet. Sheet size for testing type II machines shall be 11 by 17 inches. Sheet size for testing the binding section of type III machines shall be 8-1/2 by 13 inches. A 25 sheet lift of paper, punched along the long dimension in the corresponding size punching machine, shall be bound in the machine using a 3/16 or 7/32 inch inside diameter, narrow back binding element. In addition, a 240 sheet lift of paper, similarly punched, shall be bound in a 1-1/8 inch inside diameter, wide back binding element. After binding, the elements shall be examined, and each page of each bound group shall be laid open, one sheet at a time, to determine compliance with the applicable requirement in 3.6.

* 4.6.3 Electromagnetic compatibility tests. When electromagnetic compatibility is required for type I machines, the first article or initial production unit as applicable, shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461 (see 3.5.3.2).

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Type I machine. No packaging required.

5.1.1.2 Types II and III machines. Projecting detachable handles shall be removed from each type II and III machine in order to reduce cube and prevent damage while in transit. Alternatively, handles shall be positioned to minimize cube. Each machine shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Fiberboard inserts shall be used to fill voids within the box. The detached handle shall be secured in the same box as the machine. Each box shall be closed in accordance with method II of the appendix of PPP-B-636.

5.1.2 Level C. Machines shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I machine. Each type I machine, packaged as specified in 5.1, shall be packed in a nailed wood or cleated plywood shipping container conforming to class 2, style 2 or 4 of PPP-B-621, or overseas type of PPP-B-601. Each machine shall be cushioned, blocked, and braced or anchored, and waterproofed in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.1.2 Types II and III machines. Two type II or III machines, packaged as specified in 5.1, shall be packed in snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Type I machine. Each type I machine, packaged as specified in 5.1, shall be packed in a nailed wood or cleated plywood shipping container conforming to class 1, style 2 or 4 of PPP-B-621, or domestic type, style A or B of PPP-B-601. Each machine shall be cushioned, blocked, and braced or anchored in accordance with MIL-STD-1186.

5.2.2.2 Types II and III machines. Two type II or III machines, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

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* 5.2.2.2.1 Weather resistant container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C. Machines, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The machines covered herein are intended to be used for punching and loose-leaf, plastic strip binding of pamphlets, booklets, and similar items in offices, printing plants, and duplicating centers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type machine required (see 1.2).
(c) First article. When a first article is required (see 3.1, 4.2 and 6.4).
* (d) When electromagnetic compatibility is required (see 3.5.3.2).
(e) Applicable levels of packaging and packing (see 5.1 and 5.2).
(f) When weather resistant fiberboard shipping containers are required for level B packing (see 5.2.2.2.1).

* 6.3 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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* 6.5 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-B-43308B(GL)

BINDING MACHINE, PLASTIC BINDING ELEMENT, AND PUNCHING MACHINE, PAPER

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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